Area in Polar Coordinates

The polar area formula

1 State the formula for the area enclosed by a polar curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$.

2 Find the area enclosed by one full loop of $r = 3\cos\theta$ (i.e. $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$).

Area of a full cardioid

3 Find the total area enclosed by the cardioid $r = 1 + \cos\theta$.

Area between two polar curves

4 Find the area inside the circle $r = 3$ but outside the circle $r = 2$.

Area of one petal of a rose curve

5 Find the area of one petal of $r = 2\sin(2\theta)$, using the interval $\theta \in \left[0, \frac{\pi}{2}\right]$ where one petal is traced.

Setting up area between curves that intersect

6 Set up (but do not evaluate) the integral for the area inside both $r = 2$ and $r = 2 + 2\cos\theta$, noting they intersect where $2 = 2 + 2\cos\theta$, i.e. $\theta = \pm\frac{\pi}{2}$.

Area of a full rose curve

7 Find the total area enclosed by all 4 petals of $r = 3\cos(2\theta)$, using the fact that one petal is traced for $\theta \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$.

Area inside a limaçon

8 Find the area enclosed by the limaçon $r = 3 + \cos\theta$ (a curve without an inner loop, since the constant exceeds the coefficient).

9 Find the area of the small inner loop of the limaçon $r = 1 + 2\cos\theta$ (constant less than coefficient, so an inner loop exists), traced for $\theta \in \left[\frac{2\pi}{3}, \frac{4\pi}{3}\right]$.

Setting up area problems (AP-style, no evaluation)

10 Set up (but do not evaluate) the integral for the area inside the rose $r = 2\sin(3\theta)$ that lies in the first quadrant only, for one petal traced on $\theta \in \left[0, \frac{\pi}{3}\right]$.

11 Set up (but do not evaluate) the integral for the area between the circle $r = 5$ and the cardioid $r = 5 - 5\sin\theta$ (inside the circle, outside the cardioid), for $\theta \in [0, \pi]$.

Finding intersection points of polar curves

12 Find all points of intersection of $r = 2\cos\theta$ and $r = 2\sin\theta$ (both circles of radius 1) by setting the equations equal.

13 Explain why checking intersections of two polar curves algebraically (setting $f(\theta) = g(\theta)$) can MISS the pole as an intersection point, and why it must be checked separately.

Area using symmetry to simplify computation

14 Explain how the symmetry of $r = 3\cos(2\theta)$ about both axes allows the total area to be found by computing just one petal and multiplying, rather than integrating over the full range at once.

Area review: full worked example

15 Find the area enclosed by $r = \sqrt{\theta}$ for $\theta \in [0, \pi]$ (a spiral arc closed off by the segment back to the pole).

16 Find the area enclosed by $r = 4\sin\theta$ (a circle) for $\theta \in [0, \pi]$, and verify your answer matches the known area formula for a circle of the appropriate radius.

Comparing polar area to rectangular area (conceptual)

17 Explain why the polar area formula $A = \frac{1}{2}\int r^2\, d\theta$ uses r^2 rather than just r, by relating it to the area of a thin circular sector.

18 A student claims the area between $r = 1$ and $r = 2$ (an annulus, full circle) is simply $\frac{1}{2}\int_0^{2\pi}(2 - 1)^2\, d\theta$. Explain the error and give the correct setup.